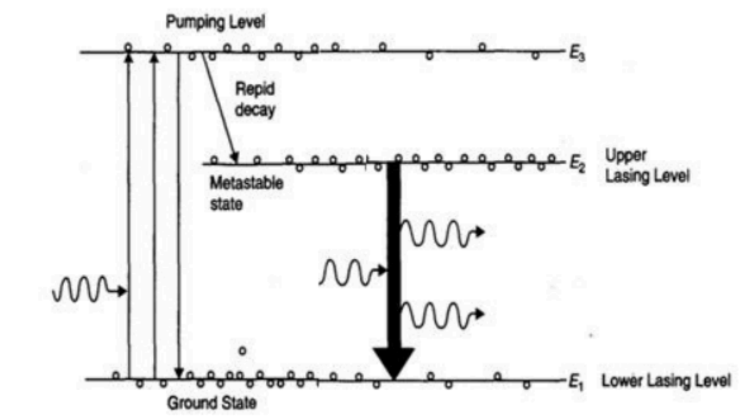
Mehul Sirsikar (roll no - 43 , reg no - 251070961)

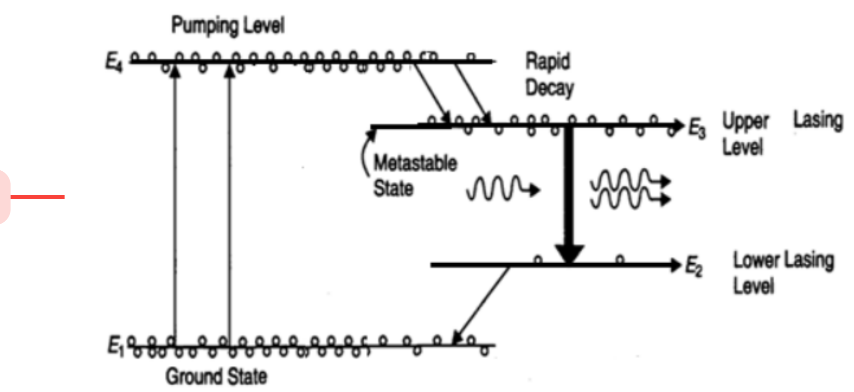
Laser

Pumping Scheme

Three level pumping scheme



Four level pumping scheme

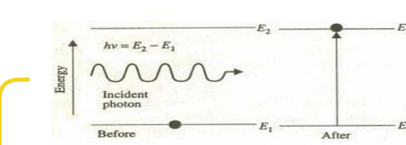


Lasing action of resonating cavity

- Pumpimg
- Population Inversion
- Light Amplification
- Spontaneous and stimulated emission
- Optical Feedback

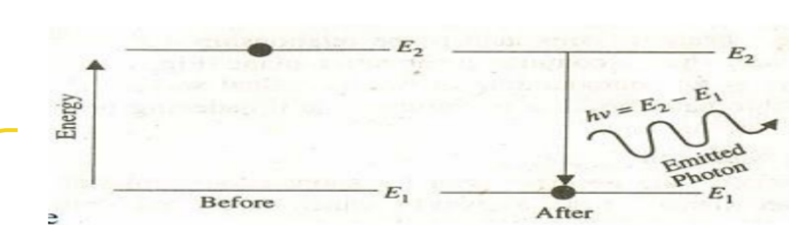
Working principle of laser

Absorption



An atom residing in the lower energy level E_1 may absorb the incident photon of energy $E_2 - E_1 = h\nu$ and jump to the excited state E_2. This transition is known as **stimulated absorption or induced absorption or simply as absorption.**

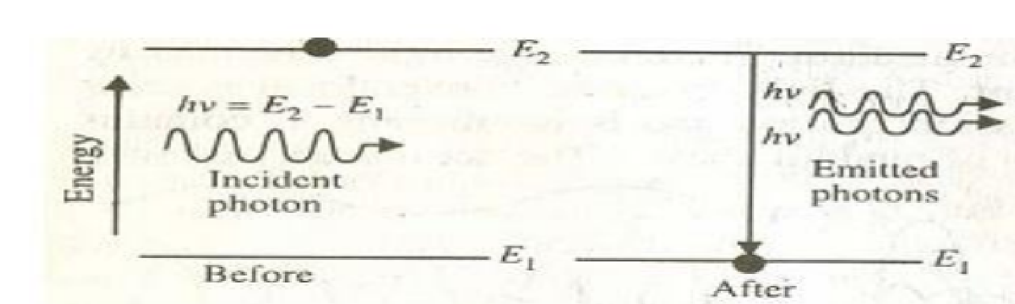
Spontaneous Emission



Excited energy state with higher energy is inherently unstable. To achieve minimum potential energy condition, atoms are always tends to remain in the lower energy level.

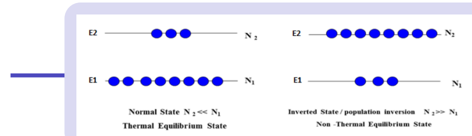
The excited atom in the state E_2 may return to lower state E_1 on its own out of natural tendency to attain minimum potential energy condition by releasing excess energy in the form of photon. This type of process in which photon emission occurs without any external impetus is called **spontaneous emission.**

Stimulated Emission



A photon of energy $h\nu = E_2 - E_1$ can *induce/ trigger* the excited atom to make a downward transition releasing the excess energy in the form of photon. The phenomenon of forced emission of photon by an excited atom due to the action of an external agency is called **stimulated emission**. It is also known as induced emission. The existence of this mechanism was predicted by Einstein in 1916. This is the principle of laser. It is represented as

Population Inversion



Population Inversion / Inverted State/ Negative Temperature State:

The number of active atoms occupying an energy state is called as **population of that state.** Let N_1 and N_2 be the population of lower energy level E_1 and upper energy level E_2. In thermal equilibrium the population of energy level E_1 and E_2 is given by Boltzmann factor

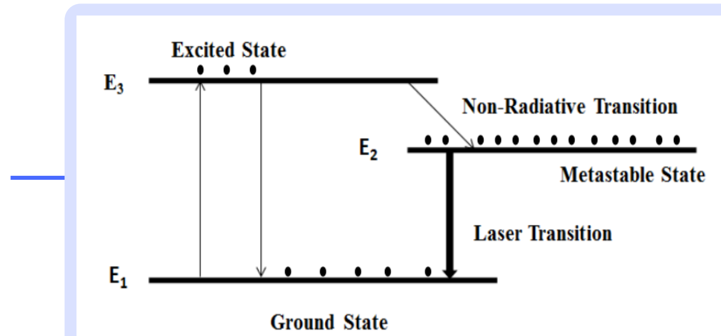
$$\frac{N_2}{N_1} = \exp\left[-\frac{(E_2 - E_1)}{kT}\right] \text{-------}(1)$$

The negative **exponent** in this equation indicates that $N_2 << N_1$ at equilibrium. It means more atoms are in lower energy levels E_1. This state is called as "Normal State" or "Thermal Equilibrium state".

Population inversion *is the condition of the material in which population of upper energy level N_2 far exceeds the population of lower energy level N_1, i.e. $N_2 >> N_1$.*

Therefore, a non-equilibrium state is to be produced in which the population of the upper energy level exceeds to a large extent than the population of lower energy level. When this situation occur the population distribution between the energy level E_1 and E_2 is said to be inverted, and medium is said to have gone into the state of population inversion.

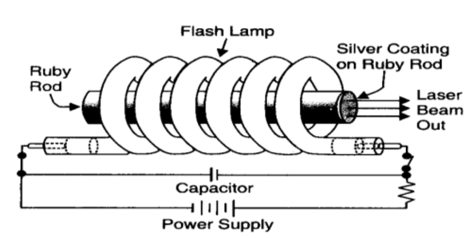
Metastable state



It is the intermediate state between ground state and excited state.

Ruby Laser

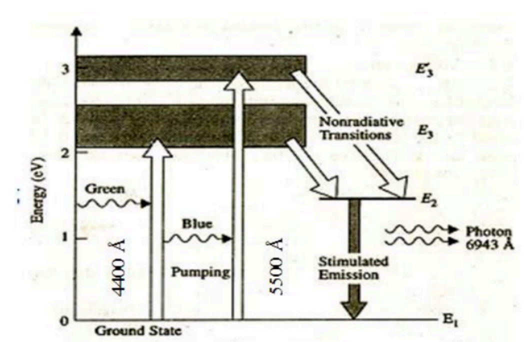
Construction



Ruby is taken in the form of cylindrical rod of about 4 cm long and 1cm in diameter. Its end faces are made perfectly flat and parallel to each other. One of them is completely silvered to achieve 100 % reflection and other one is partially silvered to make it semitransparent. Thus, two silvered faces constitute the mirrors of Fabry Perot / Optical resonator.

Ruby rod is surrounded by a helical photographic flash lamp filled with xenon. It produces flashes of white light, whenever activated by the power supply. The system is cooled with the help of coolant around the ruby rod.

Working



The energy levels of the chromium ions in the crystal are as shown in the figure 13. Ruby Laser uses three level pumping scheme. Cr^{3+} ions are the active centers while Aluminium and Oxygen atoms are inert. The xenon flash lamp generates an intense flash of white light lasting for few milliseconds. Ground state Cr^{3+} ions absorb light at pump bands 5500 Å & 4000 Å. Thus Cr^{3+} ions are excited to the energy bands E_3 and E_3' by the green and blue components of white light. The normal excited level E_3 is highly unstable which has life time of about 10^{-9} sec. From there the Cr^{3+} ions undergo non-radiative transitions and quickly drop to the metastable level E_2.